

4-bit UP/DOWN Counter using SystemVerilog Enum

Minor Project - Design, Simulation and Verification

Item	Specification
HDL	SystemVerilog
Counter width	4 bits
Direction	UP / DOWN
Special behavior	Value 4 appears for two consecutive clock cycles during DOWN counting
Reset	Synchronous, active-high
Recommended simulator	EDA Playground

Problem Definition

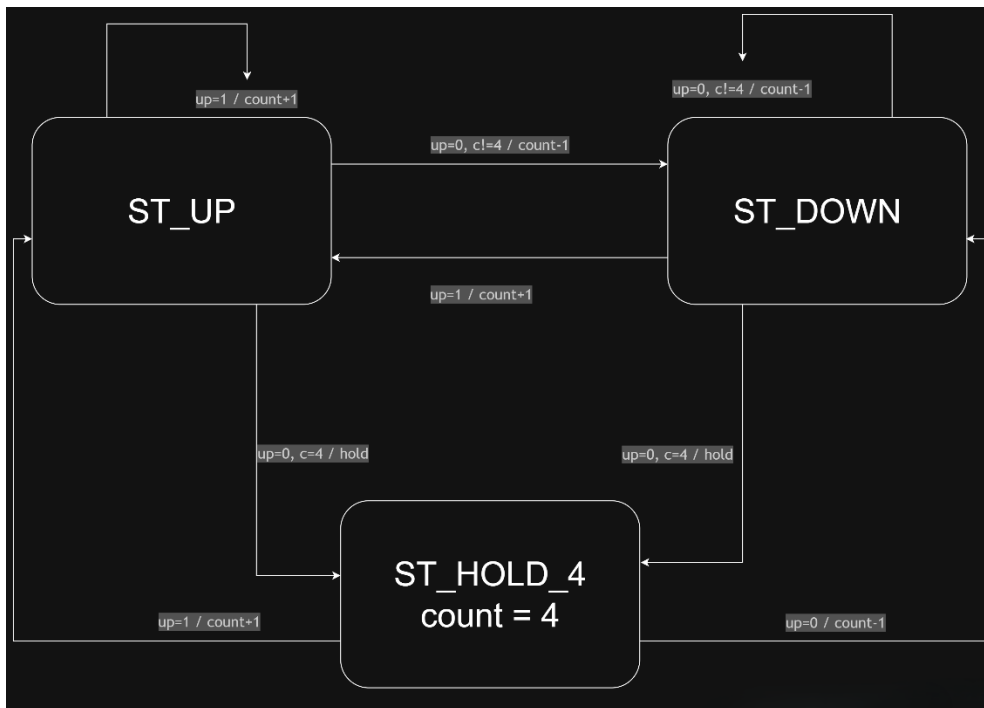
The assigned statement asks for an UP/DOWN counter in SystemVerilog using an enum datatype, with the value 4 counted twice during down-counting. Since the statement leaves the exact output sequence to the student, this project defines the requirement explicitly as **6 -> 5 -> 4 -> 4 -> 3**.

Design Approach

The counter stores a 4-bit count value. A SystemVerilog enum represents the operating state: `ST_UP`, `ST_DOWN`, and `ST_HOLD_4`. During normal DOWN counting, reaching 4 causes a transition to `ST_HOLD_4`.

The count remains 4 for that clock cycle, then the next cycle returns to `ST_DOWN` and produces 3.

State	Meaning
ST_UP	Normal upward counting
ST_DOWN	Normal downward counting
ST_HOLD_4	Hold the value 4 for one clock cycle



Functional Behavior

UP counting:

0 -> 1 -> 2 -> 3 -> 4 -> 5 -> 6 -> ... -> 15 -> 0 DOWN counting

with the required special case:

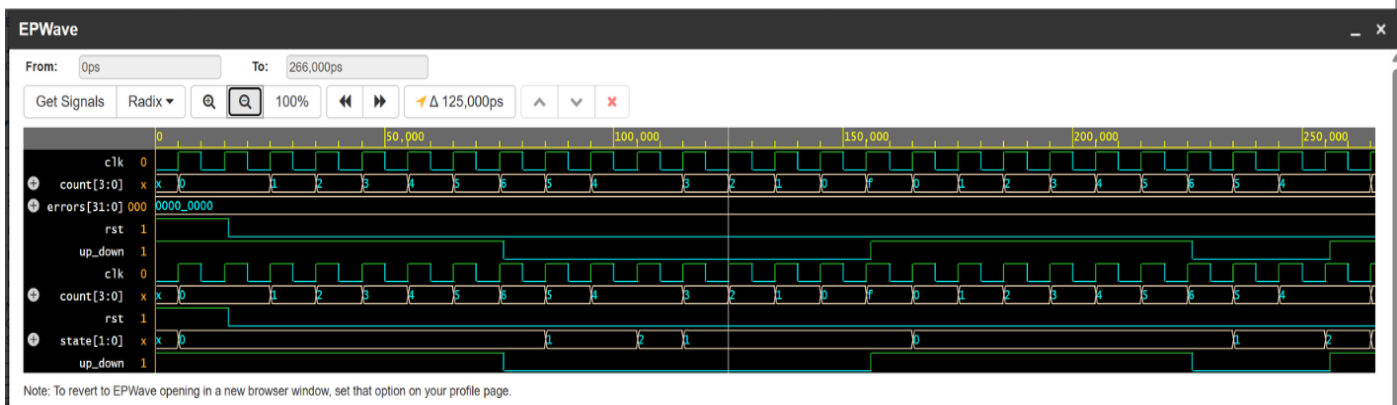
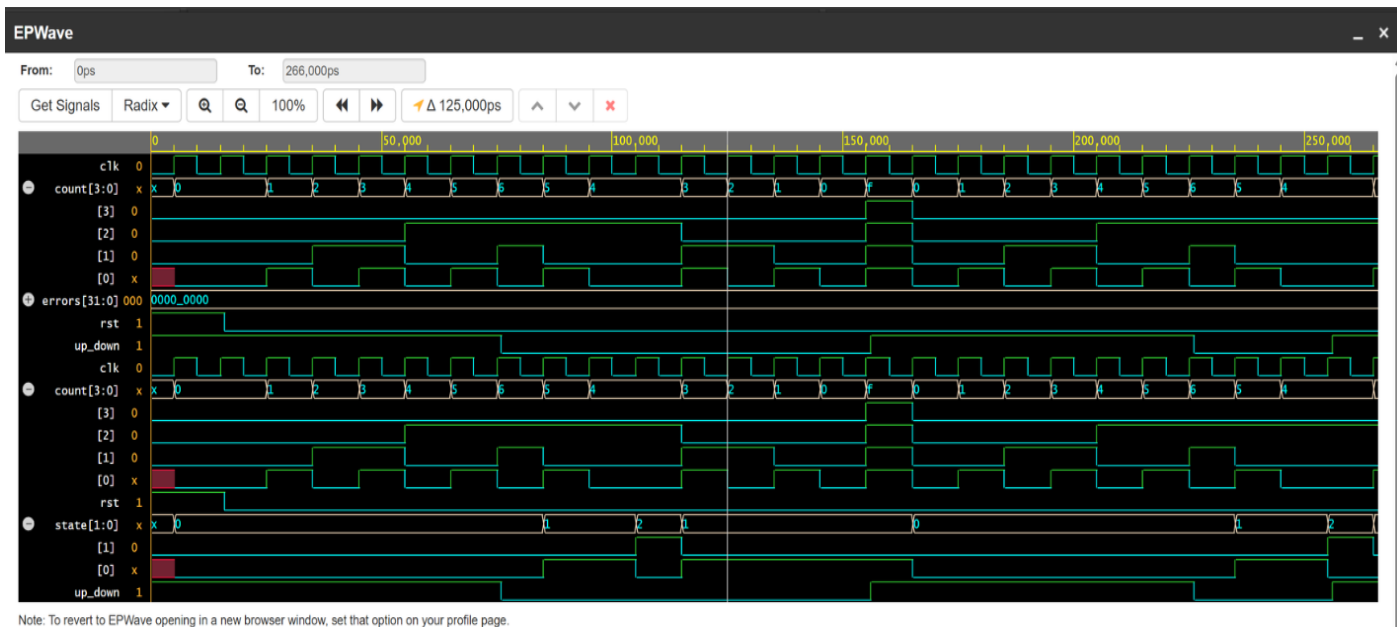
6 -> 5 -> 4 -> 4 -> 3 -> 2 -> 1 -> 0 -> 15 -> ...

The count is updated only on the rising edge of the clock. Reset is synchronous and sets the count to 0.

Verification

The testbench generates a 10 ns clock, applies reset, exercises UP counting, changes to DOWN counting, and checks the required double-4 sequence. It also checks the DOWN boundary transition 0 -> 15. Each expected value is compared with the actual output and reported as PASS or FAIL.

Test	Expected result
Reset	count = 0
UP sequence	0 -> 1 -> 2 -> 3 -> 4 -> 5 -> 6
Double-4	6 -> 5 -> 4 -> 4 -> 3
Continue DOWN	3 -> 2 -> 1 -> 0
DOWN wrap	0 -> 15



#output

[2026-09-04 15:15:58 UTC] iverilog '-Wall' '-g2012' design.sv testbench.sv && unbuffer vvp a.out

VCD info: dumpfile dump.vcd opened for output.

=== 4-bit UP/DOWN Counter Test ===

PASS: reset | count=0

PASS: UP 0->1 | count=1

PASS: UP 1->2 | count=2

PASS: UP 2->3 | count=3

PASS: UP 3->4 | count=4

PASS: UP 4->5 | count=5

PASS: UP 5->6 | count=6

PASS: DOWN 6->5 | count=5

PASS: DOWN 5->4 (first 4) | count=4

PASS: DOWN 4->4 (second 4) | count=4

PASS: DOWN 4->3 | count=3

PASS: DOWN 3->2 | count=2

PASS: DOWN 2->1 | count=1

PASS: DOWN 1->0 | count=0

PASS: DOWN 0->15 wrap | count=15
PASS: UP back to 6 | count=6
PASS: DOWN 6->5 (again) | count=5
PASS: first 4 (again) | count=4
PASS: second 4 (in ST_HOLD_4) | count=4
PASS: HOLD -> UP 4->5 | count=5
=== ALL 20 CHECKS PASSED ===
testbench.sv:81: \$finish called at 266000 (1ps)
Finding VCD file...
./dump.vcd
[2026-09-04 15:15:59 UTC] Opening EPWave...
Done

Software

Recommended: EDA Playground with a SystemVerilog-capable simulator. ModelSim/Questa or Vivado are also suitable alternatives. The project only requires simulation; FPGA hardware is not required.

Files

updown_counter.sv	RTL design
updown_counter_tb.sv	Testbench
README.md	Project notes
report.pdf	This report